

Pricing and Hedging of volatility swaps

Master 1 - Mémoire

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Abstract

The purpose of the following paper is to give an extensive overview of volatility swaps. We will take a look at their composition, the underlying mathematical components as well as their actual use in trading to give a real-life scenario for more context. Moreover, we have performed numerical analysis to benchmark the different methods encountered in literature.

Key Findings

While different models exist to simulate financial assets, the Heston model provided a solid foundation to test our benchmark. We found that the Monte Carlo method associated with Control Variate is the most efficient as well as precise method across our different tests. We highlighted how volatility swaps can be hedged using both strip of options and variance swaps in a unique relationship.

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Research questions

What are the components impacting the pricing of volatility swaps?

How to price volatility swaps from a theoretical and practical point of view and what are their possible uses for a trading desk?

Introduction

Volatility is one of the most important metrics in the finance industry because of its presence in each and every aspect or other critical measures such as expected return, Sharp ratio and so on. We have seen in recent years how volatility has gained importance on the stock market after years of a very stable economy. Since the Covid-19 and even more recently with the announcement of tariffs by Donald Trump and his “liberation day”, markets have been sent into turmoil. These events are evidence of the fact that studying volatility, hedging against it or speculating about this measure have become essential in today’s economy and it makes this master’s thesis relevant to the current market trends.

But what is volatility? Volatility can be distinguished in three ways. We have historical volatility which can be computed by aggregating data from past market dynamics, and it gives a general approximation of what we can expect as volatility for the current security. But in finance, what happened yesterday is by no means a reflection of what could happen in the future and as such, historical volatility falls short of predictive power on numerous occasions. We then have, the predicted volatility or implied volatility, this volatility is the result of the expectation of future movements of the market by countless agents. For the S&P 500, this measure is called VIX and as we will see shortly after, it can serve as a proxy for volatility if nothing else is available, but it does not represent a precise estimate. Finally, we have the current volatility of the asset which is usually only theoretically obtained but is the value that we are looking for when we want to analyze volatility.

This master’s thesis is directed towards variance and volatility swaps, and we will expand on the differences and similarities of these two securities as they share important points and differ in critical instances.

What is a variance/volatility swaps ? These securities are financial derivatives traded OTC which allows us to receive a potential payoff based on the realized (and not implied) volatility on the market. The variance swap, as its name indicates, is based on the variance of the movements observed on the market while the volatility swap is based on the volatility. They both have the same payoff structure which is $\text{Payoff} = N_{\text{var}} \times (\sigma_{\text{real}}^2 - \sigma_{\text{strike}}^2)$ for the variance swaps and $\text{Payoff} = N_{\text{vol}} \times (\sigma_{\text{real}} - \sigma_{\text{strike}})$ for the volatility swaps. From these two equations, we can deduce that both payoffs seem similar but in fact there is an important difference, the

presence of the square in the formula for the variance. This has several implications in the process of pricing both swaps and as Song-Ping Zhu and Guang-Hua Lian (2015) explained in their paper volatility swaps are thus more difficult to price because of the lack of a closed form formula contrary to variance swaps. In this paper, they also expose the different measures of volatility and their implications on the accuracy of the metrics obtained. They oppose these two

measure for a discrete sampling: $\sqrt{\frac{AF}{N} \sum_{i=1}^N \left(\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} \right)^2} \times 100$ and $\sqrt{\frac{\pi}{2NT} \sum_{i=1}^N \left| \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} \right|} \times 100$

(1) where AF is the annualized factor, N the number of observations, T the number of time periods in years and S_{t_i} is the asset price at time i. Following their recommendation, we will use the second method for a greater precision in the calculated volatility throughout our data analysis. Another method to express the volatility can be used if we suppose a continuous

sampling methodology: $\sigma = \sqrt{\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left(\frac{dS_t}{S_t} \right)^2}$. (2) Some academicians refer to this method while it faces numerous challenges, among them is the risk of error as volatility swaps are calculated through a discrete method for their payoffs as mentioned by Elliott and Lian (2012).

The objective of this master thesis is to discuss pricing methods for variance/volatility swaps as well as how to hedge them or use them in the hedging process. Indeed, hedging against the risk implied by these securities is of the utmost importance and as we will see later on, we can use replication to create both a strip of options or even variance swaps themselves. This need for replication comes from the fact that variance/volatility swaps are usually tailor made because they are traded over the counter. It is then difficult to hedge perfectly because of their illiquid and customizable nature. Replication can thus be an effective choice to hedge against the uncertainty stemming from owning such securities in our portfolios.

Mathematical Background

The aim of this chapter is to present in greater detail every mathematical aspect which might be encountered in this document. We will expose generalities about the stock market as well as the presentation of different models which will be used to then create simulations. We will define notions such as Brownian Motion, Heston model and many others which will help us gain greater insights.

Brownian motion

A Brownian Motion is a stochastic process which has the following properties:

$(B_t)_{t \geq 0}$ tel que $B_0 = 0$, $B_t - B_s \sim \mathcal{N}(0, t - s)$, independant and continuous increments

A Geometric Brownian motion is a stochastic process which is widely used in the simulation of assets' movements. It is composed of a deterministic part and of a stochastic part simulating the random part of the process. It follows this equation: $[dS_t = \mu S_t dt + \sigma S_t dB_t]$

where μ is the drift and σ the volatility of the process. The end part of the equation is the stochastic part with $[dB_t]$ being the Brownian Motion.

These equations have been used in financial markets since Louis Bachelier first decided to apply mathematics to finance with his thesis: Théorie de la spéculation (1900). This process has been the foundation of the Black-Scholes model, still widely used today to price derivatives. This model will be deepened further because it is the base of other models used to fill the gap left by Black, Scholes and Merton and the inefficiencies derived from their work.

Risk neutral measure

The risk neutral measure is an important concept to master when dealing with financial assets. It implies that investors are neutral towards risk, and it goes against the thinking of risk aversion for the sake of an easier representation. Indeed, under the risk neutral measure, every derivative is a Martingale which allows for the use of interesting properties. Usually written as Q , it states that every asset actualized has a return equal to the risk-free rate.

We have alluded to Martingales, so let's present the definition of its mathematical object.

A process (M_t) is a martingale w.r.t. a filtration $(\mathcal{F}_t)$ if: M_t is $\mathcal{F}_{t-\text{maual}}$, $E[|M_t|] < \infty$ for all t , $E[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

Semi-Markov chains

Reminder : A Markov chain is a stochastic process that is said to have no memory: the value of $X(n+1)$ only depends on the value of $X(n)$ and not on the previous iterations of $X_{i(i \in N)}$. A Markov chain is defined by its initial distribution, which is a probability distribution, and by its transition matrix.

A transition matrix called P on E is a family of reel number $P(i, j)_{(i, j) \in E^2}$ such that:

- $\forall (i, j) \in E^2, P(i, j) \geq 0$;
- $\forall i \in E, \sum_{j=1}^n P(i, j) = 1$.

This said, we can define a Markov chain: we define a Markov chain $X_n (n \geq 0)$ with an initial law μ and with a transition matrix P. Therefore, we have:

- $X_0 \sim \mu$;
- $P(X_{n+1} \in A | F_n) = P(X_n, A), P - a. s., \forall A \in F, \forall n \geq 0$ (3)

The second property here shows the absence of memory in Markov chains: the path does not really matter; the only concern is the previous state.

This said, we can now present an example of a Semi-Markov Chain (SMC):

We consider a space E, we consider $Z_n (n \geq 0)$ the state of the process at time n, $S_n (n \geq 0)$ the duration of the n-th change of state, and $J_n (n \geq 0)$ the state of the process after the n-th change of state. We also note here that $X_n = S_{n+1} - S_n$ the time spent by the process in the n-th state.

We see here that $J_n = Z_{S_n}$

If we have $P(J_{n+1} = j, X_n = k | J_0, S_0, \dots, J_n, S_n) = P(J_{n+1} = j, X_n = k | J_n)$ (4)

- (J_n, S_n) is called a Markov Renewal Chain
- Z_n is a Semi Markov Chain.

Compared to “classic” Markov chains, Semi Markov Chains are useful to represent process with variable sojourn time. In “classic” Markov Chains, sojourn time tends to grow exponentially or geometrically. The SMC process can represent an irregular evolution of sojourn time.

Mathematics for numerical implementation

We have used Monte Carlo simulation in order to price the derivatives. This technique relies on the execution of N i.i.d processes to compute the average which is supposed to be a reliable approximation:

$$\hat{I}_N = \frac{1}{N \text{ (number of simulations)}} \sum_{i=1}^N f(X_i \text{ (i.i.d. sample from } X)) \quad \text{where } f(X) \text{ is the quantity of interest}$$

(5)

Cross Validation is used to assess and find the best parameters for a model by training different models over different subsets of data and then finding the optimal parameters minimizing the function:

$$CVLoss(\theta) = \frac{1}{K \text{ (number of folds)}} \sum_{k=1}^K L^{(k)}(\theta \text{ (model parameters)})$$

Control Variate is a method we used in our Monte Carlo analysis in order to find the optimal reduction factor and as such, a more accurate simulation by reducing the number of simulations needed:

$$\widehat{I}_{CV} = \frac{1}{N \text{ (simulations)}} \sum_{i=1}^N [f(X_i) \text{ (target function)} - \varepsilon \text{ (optimal coefficient)}(g(X_i) - \mu_g \text{ (known expectation)})] \text{ and } Var(\widehat{I}_{CV}) = Var(f(X)) - 2\varepsilon Cov(f(X), g(X)) + \varepsilon^2 Var(g(X)) \quad (6)$$

The Fourier inversion technique is a widely used method in analysis to inverse a density function from its characteristic function. The objective in finance is usually to find the price of a derivative as the final result:

$$C(K, T) = S_0 \cdot P_1 - K e^{-rT} \cdot P_2 P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{e^{-iu \log K \cdot \Phi(u - i(j-1))}}{iu \cdot \chi_j(u)} \right] du \quad \text{for } j = 1, 2 \quad (7)$$

The bisection method is used in analysis to find the optimal parameters by iterating the following process under the acceptable error epsilon:

Goal: find σ^* such that $C_{BS}(S_0, K, T, r, \sigma^*) = C_{market}$

Initialize bounds $\sigma_{low}, \sigma_{high}$ $\sigma_{mid} = \frac{\sigma_{low} + \sigma_{high}}{2}$

If $C_{BS}(\sigma_{mid}) > C_{market}$, then $\sigma_{high} = \sigma_{mid}$

Else $\sigma_{low} = \sigma_{mid}$ Stop when $|\sigma_{high} - \sigma_{low}| < \varepsilon$ where ε = tolerance threshold

Out-of-Bag (OOB) Error Estimate:

For sample i , use only trees b where $i \notin$ bootstrap sample of b :

$$\widehat{f_{OOB}}(x_i) = \frac{1}{|B_i|} \sum_{b \in B_i} \widehat{f}_b(x_i), \text{ OOB error} = \frac{1}{N} \sum_{i=1}^N L(y_i, \widehat{f_{OOB}}(x_i))$$

where B_i = set of trees where i is out-of-bag, L = loss function (e.g., MSE)

Heston model

Let's take time to describe in greater detail the Heston which will play an important role in this master's thesis as most of our calculations will be based on the Heston model for the computing part. Heston (1993) presented these equations to describe financial derivatives:

$$\begin{aligned}dS_t &= \mu S_t dt + \sqrt{V_t} S_t dB_t^S \\dV_t &= \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dB_t^V\end{aligned}$$

(8)

where theta is the volatility in the long run, kappa the rapidity of this mean-reverting process, xi being the volatility of the volatility and two Brownian Motion of correlation rho. One of the great deepening by Heston compared to the Black-Scholes model (1973) is the idea of modeling volatility as a stochastic process and not like a constant overtime. This will have greater implications when we plot volatility surfaces later on. Indeed, the volatility surface of the Black Scholes model is completely flat because of the constant volatility. As such, the smile is not captured by the model. Other models like the Heston model or BN-S model try to take into account the fact that a higher implied volatility is observed for options deep in or deep out of the money.

Pricing

To understand the pricing formulas underlying the Heston model, it is needed to present the two types of solutions we can find for a given problem.

The first type of these solutions is what we call a semi-closed form formula which as we will see later on is an equation with a non-trivial component and requires further numerical computation to obtain a number as a result. The second type of solution we can find is what we call a closed form formula, which is completely free of this restriction of further computation. They are nonetheless harder to find as they often require further research and sometimes ingenious perspective to deduce a result.

From there, the Heston model usually offers two perspectives for the same problem, the Fourier Transform techniques or the PDE approach.

While the PDE approach is often numerically more intensive and gives more often than not a semi-closed formula, Zhu and Lian (2010) managed to deduce a closed form solution to this problem. They rely on the Heston model with risk neutral modified parameters and this definition of volatility: $\sigma_R^2 = \frac{AF}{N} \sum_{i=1}^N \left(\frac{S_i - S_{i-1}}{S_{i-1}} \right)^2 \times 100^2$. In most cases, the complexity to find a solution to this problem is the mean of the two stochastic variables taking different forms depending on the definition of the volatility that we use. They then introduce a new variable

$$I_t = \int_0^t \delta(q_{s-1} - r) S_s ds. \text{ to solve the following PDE : } \frac{\partial U_i}{\partial t} + \frac{1}{2} S^2 \frac{\partial^2 U_i^2}{\partial S^2} + \rho \sigma_r S \frac{\partial^2 U_i^2}{\partial S \partial r} + \frac{1}{2} \sigma_r^2 \frac{\partial^2 U_i^2}{\partial r^2} + r S \frac{\partial U_i}{\partial S} + [x^*(w^* - v)] \frac{\partial U_i}{\partial r} - r U_i + h(q_{i-1} - r) \frac{\partial U_i}{\partial I} = 0. \quad (9)$$

The terminal condition is

$$U_i(S, r, I, t_i) = \left(\frac{S}{I} - 1 \right)^2.$$

The final objective of this process is to decompose the two-dimensional problem in two one dimensional problems using the technique pioneered by Little and Pant (2001). They rely on this consequence of the Feynman-Kac theorem: $E_0^Q \left[\left(\frac{S}{I_i} - 1 \right)^2 \right] = e^{rt} U_i(S, v_0, I_0, 0)$. to

reduce the PDE to this state : $\frac{\partial U_i^2}{\partial t} + \frac{1}{2} r S^2 \frac{\partial U_i^2}{\partial S^2} + \rho \sigma_r S \frac{\partial U_i^2}{\partial S \partial r} + \frac{1}{2} \sigma_r^2 \frac{\partial U_i^2}{\partial r^2} + r S \frac{\partial U_i}{\partial S} + [x^*(w^* - v)] \frac{\partial U_i}{\partial v} - r U_i = 0$. (10) From there, they used the Fourier Inverse methodology to simplify the

final solution to $K_{\text{var}} = E_0^Q [\sigma_R^2] = \frac{e^{r\Delta t}}{T} [f(v_0) + \sum_{i=2}^N f_i(v_0)] \times 100^2$. This paper allows us to really demonstrate the diversity of methodology possible to solve the same existing problem caused by the two stochastic parts in the mean symbol of the original equation. It relies on a PDE approach and nonetheless uses the Fourier Inverse method to then deduce the fair strike which allows us to price the volatility swap.

Another approach all together despite its reliance on PDE is done by using regime switching to simulate a Markov Chain with the different states of the economy as done by Elliott, Siu and Chan (2006) and analytically solving the resulting PDE instead of using a reduction technique. Modeling a Markov chain is a useful tool to simulate macroeconomics events and take them into account into the pricing of derivatives as the different states of the chain represent the

different economic conditions possible. They rely on a general Stochastic Volatility model of which the Heston model is a special case and a Markov chain. They used Esscher Transform to deduce a martingale measure in an incomplete market as an infinity exists in these conditions. They mentioned the drawbacks of this method which consists in non-linearity, something usually sought after by academicians to price derivatives. They define the derivative

$$\frac{dQ^\Theta}{dP} \Big|_{\mathcal{G}_t} = \exp \left(\int_0^t \Theta_u \sigma_u dW_u^1 - \frac{1}{2} \int_0^t \Theta_u^2 \sigma_u^2 du \right).$$

with $\mathcal{G}_t$ being the sigma algebra, θ the regime switching Esscher Process and Q the Esscher transform. This equation is the foundation used to find the values of the central elements of the pricing equation:

$$E_{Q_s} \left[e^{-\int_t^T r_s du} \sigma_t^2 \mid \Sigma, V_t = V, X_t = X \right] \text{ and } E_{Q_s} \left[e^{-\int_t^T r_s du} \sigma_{R,T}^2(S) \sigma_t^2 \mid \Sigma, V_t = V, X_t = X \right].$$

They described the first element as being a zero-coupon bond and they used PDEs to decompose the second element. While this paper gives a semi closed form formula for the variance swap, the volatility swap result is not so clear as the second element is only described as a resulting PDE and the authors only mentioned that the last step was to solve this equation to find the fair strike of the volatility swaps without going further as to solve it in the paper. This issue is what we referred to when we mentioned the difficulty of finding analytical solutions for PDEs. Without the use of simplification techniques, finding a semi closed form formula using PDEs is more difficult than other methods.

The Fourier Transform method is another technique used to deduce a formula for the pricing of the volatility swap. It induces the use of the characteristic function and the Fourier Inverse to then deduce a simplified result of the fair strike of a volatility swap. The most important methods used to numerically find a result are the Carr-Madan methodology and the COS technique as we will see in the section. The main difference between those two techniques relies on the transcription of the problem from a numerical standpoint. The Carr-Madan technique relies on the use of the Fast Fourier Transform technique while the COS method uses cosine series to approximate the option using Fourier cosine expansion.

Zhu and Lian (2015) perfectly demonstrate the application of the Carr-Madan method, albeit modified to fit the current problem they faced. Indeed, they used the same principle by using *the characteristic function but more precisely the forward characteristic function given by :*

$$f(\phi; T, t, V_0) = e^{C(t, T-t)} g(D(\phi, T-t); T, V_0)$$

where

$$C(\phi, \tau) = r\phi\tau + \frac{\kappa^Q \theta^Q}{\sigma_V^2} \left[(a+b)\tau - 2 \ln \left(\frac{1-ge^{b\tau}}{1-g} \right) \right]$$

$$D(\phi, \tau) = \frac{a+b}{\sigma_V^2} \frac{1-e^{b\tau}}{1-ge^{b\tau}}, a = \kappa^Q - \rho\sigma_V\phi, b = \sqrt{a^2 + \sigma_V^2(\phi - \phi^2)}, g = \frac{a+b}{a-b} \text{ and}$$

$$g(\phi; \tau, V) = \exp \left(\frac{-2\kappa^Q}{\sigma_V^2} \ln \left(1 + \frac{\sigma_V^2 \phi}{2\kappa^Q} (e^{-\kappa^Q \tau} - 1) \right) + \frac{2\kappa^Q \phi}{\sigma_V^2 + (2\kappa^Q - \sigma_V^2)\phi e^{\kappa^Q \tau}} V \right). \quad (11)$$

From there, they extracted the probability density function of the variable $\ln(S_t i) - \ln(S_{t-1} i)$ using the inverse Fourier transform of the above function. The summation of the numerous terms composing the formula for the fair strike as a consequence of using a discrete sampling variance gives the formula:

$$K_{vol} = E_0^Q[RV_{d2}(0, N, T)] = \sqrt{\frac{\pi}{2NT}} \sum_{i=1}^N \left| \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} \right| \times 100 =$$

$$\sqrt{\frac{2}{\pi NT}} \int_0^\infty \sum_{i=1}^N \text{Re} \left[\frac{f(\phi j + 1; t_i, t_{i-1}, V_0) - f(\phi j; t_i, t_{i-1}, V_0)}{\phi j} \right] d\phi \times 100. \quad (12)$$

While not using the FFT technique in the end of the paper, this is only because their point was deducing a closed form formula for volatility swap without the need to actually price on real data except by using Monte Carlo simulation.

An interesting result highlighted by Elliott and Lian (2012) is the exact same formula as described precedingly to price a volatility swap by using a regime switching model using a Markov chain to simulate different economic states. They used the same methodology as the Forward Characteristic Function and the Fourier Transform but based on an initial different assumption.

To get a proper sense of the FFT and the COS technique we have sought the paper of Lian, Chiarella and Kalev (2014) where they give an explanation of both methods through their work. The idea of the Fourier inverse technique is to find the solution of this integral where $h(x)$ is the payoff function of the security (square root of RV in the case of a volatility swap):

$$E[h(x)] = \frac{1}{2\pi i} \int_{\phi_r - i\infty}^{\phi_r + i\infty} H(\phi) E[e^{\phi x}] d\phi,$$

Nonetheless, the numerical calculation of this integral can be complex, which is why the Fourier Cosine series expansion is an efficient alternative. It usually relies on an approximation of the actual price albeit precise but an approximation still. Using the same notation as above the

general for a volatility derivatives that they used was $\sum_{j=0}^J A_j \text{Re} \left[\int_a^b h(x) \exp \left(j\pi i \frac{x-a}{b-a} \right) dx \right]$ (13). The truncated norm from infinity to a countable number J is possible because of the decay of the last terms. This pricing formula can be used as they did to accurately price volatility swaps.

In the end, different methods exist to price a volatility swap under the Heston framework. The first difference is whether we use a PDE or the Fourier Transform. Both use the Fourier inverse but differ in methodology and difficulty. If we choose the Fourier Transform, we can choose from either the Carr-Madan method or the COS technique for numerical applications depending on whether we prefer a cosine series expansion or the Fast Fourier Transform. On top of this plethora of approaches, choosing to represent the problem using regime switching might add another layer of complexity to the pricing process and its integration in the various other methodologies.

BN-S Model

From the work of Salmon and SenGupta.I (2021), on its fundamental approach, this model aims to deal with big jumps in volatility, a point where many models struggle to be precise. The model is based on a way to compute volatility based on Ornstein-Uhlenbeck type stationary process for stochastic volatility of log-returns, where the subordinator is a Lévy process with positive increments (instead of a Gaussian component).

We need to introduce here some definitions to understand the construction of the model and how it works.

- We will consider here a multivariate Gaussian random variable $X: \Omega \rightarrow R^n$ with mean μ and covariance matrix Σ .
- A Gaussian process is a stochastic process where all finite-dimensional distributions are Gaussian. It is defined by two functions, called mean and covariance:

1. $\mu(t) = E(X_t)$

2. $\rho(s, t) = E(\langle X_s - \mu(s), X_t - \mu(t) \rangle)$

- Two Gaussian processes have the same law $\Leftrightarrow$ mean and covariance functions are the same.
- Let $0 < H \leq 1$. By theorem, there exists a continuous centered Gaussian process (its mean function equals 0) W_t^H with covariance function given by

$$\rho^H(s, t) = \frac{1}{2(|t|^{2H} + |s|^{2H} - |t - s|^{2H})}. \quad (14)$$

Its sample paths are α Hölder for all $0 < \alpha < H$.

- Definition: Let $g(t)$ be $\alpha > 1/2$ Hölder continuous. Let $\sigma(t)$ be C^2 with bounded derivatives and $\mu(t)$ Lipschitz. We say $x(t)$ solves the differential equation

$$dx(t) = \sigma(x(t))dg(t) + \mu(x(t))dt$$

In the Young sense if

$$x(t) - x(0) = \int_0^t \sigma(x(u))dg(u) + \int_0^t \mu(x(u))du, \quad (15)$$

where we assume a valid solution $x(t)$ is $\beta > 1-\alpha$ Hölder so that the integrals are well defined Young integrals.

- Suppose that G_t is a.s. $\alpha > 1/2$ Hölder. Then we say that X_t solves the differential equation

$$dX_t = \sigma(X_t)dG_t + \mu(X_t)dt,$$

In the Young sense if for almost all $\omega \in \Omega$ we have that $X_t(\omega)$ solves the differential equation in the Young sense.

On a fractional log-variance model, we'll consider here in our model $H > 1/2$, and consider the corresponding Brownian Motion with $W=0$ a.s.. We note that W is α Hölder for every $0 < \alpha < H$, and since $H > 1/2$, we have in the Young sense

$$\sigma_t^2 = e^{X_t}, dX_t = \alpha(m - X_t)dt + v dW_t^H$$

We'll now consider two lemmas useful for the following:

- The process $X_t = e^{-\alpha t}(X_0 - m) + m + vW_t^H - \alpha \int_0^t e^{-\alpha(t-s)} vW_s^H ds$ solves

$$dX_t = \alpha(m - X_t)dt + v dW_t^H \quad \text{in the Young sense. (17)}$$

- For any continuous Gaussian process $X_t \in R$ with covariance function $\rho(s, t)$ and $\mu(t) = E(X_t)$, $Y_t = \int_0^t X_s ds$ is Gaussian with covariance $\int_0^t \int_0^s \rho(x, y) dx dy$ and mean $\int_0^t \mu(x) dx$. (18)

The second lemma mentioned just above tells us that $-\alpha \int_0^t e^{-\alpha(t-s)} vW_s^H ds$ is Gaussian with mean 0 and covariance $(av)^2 \int_0^t \int_0^s e^{(-\alpha((s-x)+(t-y)))} \rho^H(x, y) dx dy$. (19)

We also can see that $e^{-\alpha t}(X_0 - m) + m + vW_t^H$ is Gaussian with mean $e^{-\alpha t}(X_0 - m) + m$ and covariance $\rho^H(s, t)$. We need now a way to obtain the covariance between $e^{-\alpha t}(X_0 - m) + m + vW_t^H$ and $-\alpha \int_0^t e^{-\alpha(t-s)} vW_s^H ds$ to have a complete description of X_t .

- By lemma, for any Gaussian process X_t , we have $E \left[X_s \int_0^t \llbracket X_r dr \rrbracket \right] = \int_0^t \rho(s, x) - \mu(s)\mu(x)dx.$
- By theorem, the process $X_t = e^{-\alpha t}(X_0 - m) + m + vW_t^H - \alpha \int_0^t e^{-\alpha(t-s)}vW_s^H ds$ is a Gaussian process with mean $e^{-\alpha t}(X_0 - m) + m$ and covariance function $\rho(s, t) = v^2 \left(\rho^H(s, t) - \alpha \left(\int_0^t e^{-\alpha(t-x)}\rho^H(s, x)dx + \int_0^s \llbracket e^{-\alpha(s-x)}\rho^H(t, x)dx \rrbracket + \alpha^2 \int_0^t \int_0^s \llbracket e^{-\alpha((s-x)+(t-y))}\rho^H(x, y)dxdy \rrbracket \right) \right]$
- Let W_t^H be a centered fractional Brownian motion with $H > 1/2$, and consider the fractional Ornstein-Uhlenbeck process $dX_t = \alpha(m - X_t)dt + v dW_t^H$. Since X_t is Gaussian with mean $e^{-\alpha t}(X_0 - m) + m$ and covariance $\rho(s, t)$, we have $E(e^{X_t}) = e^{e^{-\alpha t}(X_0 - m) + m + \frac{1}{2}v(t)}$, and so we have

$$E \left[\frac{1}{T} \int_0^T \llbracket e^{X_t} dt \rrbracket \right] = \frac{1}{T} \int_0^T e^{e^{-\alpha t}(X_0 - m) + m + \frac{1}{2}v(t)} dt$$

with

$$v(t) = v^2 \left(\rho^H(t, t) - 2\alpha \int_0^t e^{-\alpha(t-x)}\rho^H(t, x)dx + \alpha^2 \int_0^t \int_0^t \llbracket e^{-\alpha((t-x)+(t-y))}\rho^H(x, y)dxdy \rrbracket \right) \quad (20)$$

For the calculation of the volatility, on this model it is based on past realized volatility, adapted to a new stochastic model.

The realized volatility of the asset during the lifetime of the contract is $\sigma_R = \sqrt{\frac{1}{T} \int_0^T \sigma_d^2 dt}$

The basic formula for pricing a volatility swap is given by $P_{Vol} = E^Q[e^{-rT}(\sigma_R - K_{Vol})]$.

Approximation of the expected volatility:

$$E^Q[\sigma_R] = E^Q \left[\sqrt{\sigma_R^2} \right] \approx \sqrt{E^Q[\sigma_R^2]} - \frac{Var^Q[\sigma_R^2]}{8(E^Q[\sigma_R^2])^{\frac{3}{2}}} \quad (21)$$

This estimate is not effective for big fluctuations in volatility, which justify the will to modify it.

- BN-S model

We consider here a riskless asset with return rate r and a stock are traded until a date T . We assume the stock price is equal to $S_t = S_0 e^{X_t}$, with:

$$\begin{aligned} dX_t &= (\mu + \beta \sigma_t^2)dt + \sigma_t dW_t + \rho dZ_{\lambda t}, \\ d\sigma_t^2 &= -\lambda \sigma_t^2 dt + dZ_{\lambda t}, \sigma_0^2 > 0 \end{aligned} \quad (22)$$

$\mu, \beta, \rho, \lambda$ in $\mathbb{R}, \lambda > 0$ and $\rho \leq 0$. The arbitrage free price of the variance swap is given by

$$P_{Var} = e^{-rT} \left[\frac{1}{T} (\lambda^{-1} (1 - e^{-\lambda T}) (\sigma_0^2 - \kappa_1) + \kappa_1 T) + \rho^2 \lambda \kappa_2 - K_{Var} \right], \quad (23)$$

With κ_1 and κ_2 being respectively the expected value and the variance of Z_1 . The process $Z = Z_{\lambda t}$ is here as a subordinator. $W = (W_t)$ is a Brownian motion. W and Z are independent here.

- Fractional BN-S Model :

On this model we also have assumptions of a frictionless financial market, with a riskless asset with a return rate r and a stock traded until date T . We also have :

Derivative price: $S_t = S_0 e^{Y_t}$,

$$dY_t = b_t dt + \sigma_t dW_t + \rho dZ_t, b_t = (c_1 - c_2 \sigma_t^2),$$

$$d\sigma_t^2 = \lambda(e^{Y_t} - \sigma_t^2)dt + e^{X_t} dX_t + dZ_t,$$

where $dX_t = \alpha(m - X_t)dt + v dW_t^H, Z_0 = 0$.

c_1 and $c_2 > 0, \rho \leq 0$. W is a standard Brownian motion, W_t^H is a fractional Brownian motion with $H > 1/2$, and the process Z is a subordinator. The fact that $H > 1/2$ states that the model is focused on the long-term dependance.

We observe that $e^{\lambda t} (d\sigma_t^2 + \lambda \sigma_t^2 dt) = \lambda e^{\lambda t + X_t} dt + e^{\lambda t + X_t} dX_t + e^{\lambda t} dZ_t$ and $d(e^{\lambda t} \sigma_t^2) = d(e^{\lambda t + X_t}) + e^{\lambda t} dZ_t$. Therefore,

$$\sigma_t^2 = m_t + e^{X_t} + \int_0^t e^{-\lambda(t-s)} dZ_s, \text{ where } m_t = e^{-\lambda t} (\sigma_0^2 - e^{X_0}).$$

For the Fractional BN-S stochastic volatility model,

$$E^Q(\sigma_R^2) = \frac{v_1(T)}{T} + \frac{1}{T} \int_0^T e^{e^{-\alpha t}(X_0 - m) + m + \frac{1}{2}v(t)} dt \quad (24)$$

where $v(t)$ is given above, and $v_1(T) = \frac{1}{\lambda((1-e^{-\lambda T})(\sigma_0^2 - e^{X_0}) + (T - \lambda^{-1}(1-e^{-\lambda T}))\kappa_1)}$.

And from that, the arbitrage free price of the variance swap for the rough BN-S stochastic volatility model is $P_{Var} = \frac{e^{-rT}}{T} \int_0^T e^{e^{-\alpha t}(X_0 - m) + m + \frac{1}{2}v(t)} dt + \mu_1(T)$, where

$$\mu_1(T) = e^{-rT} \left(\frac{v_1(T)}{T} - K_{Var} \right).$$

We will now focus on the pricing of volatility swaps. We add here another assumption: $\sigma_0^2 = e^{X_0}$, justified by the idea that the process does not start right after a volatility jump. With this assumption, if we have $X_0 = m$, the process will get back to its mean at a rate of α . This assumption is thus nonrestrictive, this assumption starts the process away from $\exp(m)$, which is caused by the continuous drift of e^{X_t} rather than a jump in σ_t . We therefore assume that $m_t = 0$, and consequently $\sigma_t^2 = e^{X_t} + \int_0^t e^{(-\lambda(t-s))} dZ_s$.

We observe that $E^Q \left[\left(\frac{1}{T} \int_0^T \mathbb{E}[e^{X_t} dt] \right)^n \right] = \left(\frac{1}{T^n} \right) \int_{[0,T]^n} E^Q[e^{X_{t_1} + \dots + X_{t_n}}] dt_1 \dots dt_n$. Since the sum of the X_{t_i} is Gaussian and we know its mean and covariance functions, it leads us to

$E^Q[e^{X_{t_1} + \dots + X_{t_n}}] = e^{\sum_{i=1}^n \mu(t_i) + \frac{1}{2} \sum_{i,j=1}^n \rho(t_i, t_j)}$, which finally leads to

$$E^Q \left[\left(\frac{1}{T} \int_0^T \mathbb{E}[e^{X_t} dt] \right)^n \right] = \left(\frac{1}{T^n} \right) \int_{[0,T]^n} e^{\sum_{i=1}^n \mu(t_i) + \frac{1}{2} \sum_{i,j=1}^n \rho(t_i, t_j)} dt_1 \dots dt_n. \quad (25)$$

We introduce $\beta > 0$ such that $\sigma_R^2 < 2\beta^2$ (possible for any set of empirical financial data. We assume this, and with $\sigma_0^2 = e^{X_0}$, the arbitrage free value of the volatility swap is given by:

$$P_{Vol} = e^{(-rT)} \left(\beta \sum_{n=0}^{\infty} \frac{(-1)^{n+1} (2n)!}{(4^n (n!)^2 (2n-1))} \frac{1}{\beta^{2n} \sum_{k=0}^n A_{n-k} B_k - K_{Vol}} \right) \quad (26)$$

Where $A_k = E^Q \left[\left(-\beta^2 + \frac{1}{\lambda T} \int_0^{\lambda T} \mathbb{E}[(1 - e^{-s}) dZ_s] \right)^k \right]$ and $B_k = E^Q \left[\left(\frac{1}{T} \int_0^T \mathbb{E}[e^{X_t} dt] \right)^k \right]$. for any k .

The models we developed here (BN-S model and Fractional BN-S model, which is derived from the first one) are both based on two properties that are very important to obtain efficient

pricing, the long-term memory and the jumps. The positive correlation in the fractional Brownian motions ($H > 1/2$) is the key element that helps us here. We ended up with formulas for both variance and volatility swaps.

Local volatility model

As described by Ben Salah, Mancin and Sanhaji (2018), the model is focused on short term volatility swaps, that are less affected by stochastic interest rates, but way more sensitive to changes in volatility caused by important events.

For the model we consider the probability space $(\Omega, \mathcal{F}, P)$ and we define T as a time horizon, a filtration $\mathcal{F} = (\mathcal{F}_t)_{t \in [0, T]}$, W a standard $\mathcal{F}$ -Brownian motion and a continuous Markov chain $\xi(t): [0, T] \rightarrow \{1, 2, \dots, n\}$ taking $n \in \mathbb{N}$ many states. We assume FX rate follows a Local Stochastic Volatility Markov Switching Model, which combines several useful tools, including local volatility (volatility depends on the actual level of the underlying), Stochastic volatility and Markov Switching, which helps to modelize different market contexts, and satisfies

$$\frac{dS_t}{S_t} = (r_d - r_f)dt + A(t, S_t)\sigma_{\xi(t)}dW_t \quad (27)$$

where $(t, x) \rightarrow \sigma_{t,x}, (t, y) \rightarrow A(t, y)$ are deterministic functions, and r_d and r_f are the constant domestic and foreign interest rate respectively. We also assume

$$\sigma_{t,x} = e^{(x-1-\lfloor \frac{n}{2} \rfloor)\alpha(t)}$$

where α is bounded and deterministic function that describes the magnitude of the volatility of volatility of the sport rate (referred as vol-of-vol).

The aim of the model is to price volatility swaps, and especially estimate the term C in the following expression:

$$E \left[\sqrt{\sum_{i=1}^N \left(\log \left(\frac{S_{T_i}}{S_{T_{i-1}}} \right) \right)^2} \right] = CE \left[\sqrt{\sum_{i=1}^N E \left[\left(\log \left(\frac{S_{T_i}}{S_{T_{i-1}}} \right) \right)^2 \middle| \mathcal{F}_{T_{i-1}} \right]} \right]$$

C here represents a constant correction factor.

We need here to introduce the random variable $B^i = \sum_{j=1}^i f(S_{T_{j-1}}, \xi(T_{j-1})) =$

$$E \left[\left(\left(r_d - r_f - \frac{1}{2} \sigma_{\xi(T_{i-1})}^2 A^2(T_{i-1}, S_{T_{i-1}}) \right) \Delta T_i + \sigma_{\xi(T_{i-1})} A(T_{i-1}, S_{T_{i-1}}) \sqrt{\Delta T_i} (W_{T_i} - W_{T_{i-1}}) \right)^2 \middle| \mathcal{F}_{T_{i-1}} \right] \quad (28)$$

with $\Delta T_i = T_i - T_{i-1}$.

So to say it simply, B^i is the sum of the volatilities observed up to time i .

We can already approximate the price of the volatility swap:

$$E[\sqrt{V}] \approx CE[\sqrt{B^N}] = CE\left[\sqrt{B^{N-1} + f\left(S_{T_{N-1}}, \xi(T_{N-1})\right)}\right] = CE\left[g\left(B^{N-1}, S_{T_{N-1}}, \xi(T_{N-1})\right)\right]$$

for a function g to be found.

We already have a trivial relationship for the B^i , which is $B_m^i = B_m^{i-1} + f(S_{T_{i-1},j}, \xi(T_{i-1}), j > 1, r = 1, \dots, n$.

We have in this framework $C = \frac{E\left[\sqrt{\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 A^2(T_{i-1}, S_{T_{i-1}}) G_i^2 \Delta T_i}\right]}{E\left[\sqrt{\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 A^2(T_{i-1}, S_{T_{i-1}}) \Delta T_i}\right]}$ (29), where the G_i are

independent samples of Gaussian distributions. Here we have to choose in assumptions to pursue the calculations. If we consider a flat volatility term structure, which implies FX returns

to be normally distributed with no drift and constant volatility, we have $C = \frac{E\left[\sqrt{\sum_{i=1}^N \sigma^2 G_i^2}\right]}{\sqrt{\sum_{i=1}^N \sigma^2}}$, which

leads further to $C = 1 - \frac{1}{4N}$.

The problem of the estimation of C remains, as we may want to be more precise (the value here is a rough estimate, that is only based on simple statistics and extended to its mathematical limit). To be able to take in account every possible variation, every event that could alter the value of the swap, we will consider the whole volatility term structure.

We want here to compute $\frac{E\left[\sqrt{\sum_{i=1}^N \sigma_i^2 G_i^2}\right]}{\sqrt{\sum_{i=1}^N \sigma_i^2}}$ where σ_i is a constant volatility model ATM. Several

approaches can be considered to compute the numerator.

1) Lognormal mapping:

Firstly, let's explore the idea with a lognormal moment matching technique. We approximate $E[\sqrt{V}]$ with $E[\sqrt{v}]$, with $\ln(v) \sim \mathcal{N}(E[V], \text{Var}(V))$:

$$E[\sqrt{V}] \approx \sqrt{E[V]} \exp\left(-\frac{1}{8} \log\left(\frac{E[V^2]}{E[V]^2}\right)\right)$$

$$\approx \sqrt{E[V]} \exp\left(-\frac{1}{8} \log\left(\frac{\text{Var}(V)}{E[V]^2} + 1\right)\right) \quad (30)$$

Since we have for any i $\sigma_i \Delta T_i$ where $\sigma_{avg} = \frac{1}{N} \sum_{i=1}^N \sigma_i$ and $(\omega_i)_{i=1, \dots, N}$ are suitably defined weights,

we end up with $C = \frac{E[\sqrt{V}]}{\sqrt{E[V]}} \approx \exp\left(-\frac{1}{8} \log\left(\frac{2 \sum_{i=1}^N \omega_i^4}{(\sum_{i=1}^N \omega_i^2)^2} + 1\right)\right)$, with N growing to the infinite. If we don't have changes from the interest rates, the model converges to Black-Scholes, and the good strike of the volatility swap would be the ATM volatility. This correction term is efficient for long maturities, but not for short-term (which is the goal of our process here).

2) Numerical integration:

We use here the Laplace transformation of $\sum_{i=1}^N \sigma_i^2 G_i^2 \Delta T_i$, which is an approximation of $\sum_{i=1}^N \left(\log\left(\frac{S_{T_i}}{S_{T_{i-1}}}\right)\right)^2$. We therefore have

$$\begin{aligned} E\left[\exp\left\{-\psi \sum_{i=1}^N \left(\log\left(\frac{S_{T_i}}{S_{T_{i-1}}}\right)\right)^2\right\}\right] &= E[\exp\{-\psi \sum_{i=1}^N \sigma_{i,new}^2 G_i^2\}] \\ &= \prod_{i=1}^N E[\exp\{-\psi \sigma_{i,new}^2 G_i^2\}] \\ &= \prod_{i=1}^N \frac{1}{(1 + 2\psi \sigma_{i,new}^2)^{\frac{1}{2}}} \end{aligned}$$

Which leads to $E[\sqrt{V}] = \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{1 - E[e^{-\psi V}]}{\psi^{\frac{3}{2}}} d\psi$, with $\sigma_{i,new}^2 = \sigma_i^2 \Delta T_i$. (31)

3) WoodF:

This method is another moment matching technique, this time with the F-distribution. We need here

$$\begin{aligned} \alpha_1 &= 2k_1(k_1 k_3 + k_1^2 k_2 - k_2^2), \\ \alpha_2 &= 3 + 2k_2(k_2 + k_1^2) r_2, \text{ with} \\ r_1 &= k_1 k_2^2 + k_3(k_2 - k_1^2), \end{aligned}$$

$$r_2 = k_1 k_3 - 2k_2^2$$

Obtained from the moments of $\sum_{i=1}^N \sigma_i^2 G_i^2 \Delta T_i$.

Therefore $\sum_{i=1}^N \sigma_i^2 G_i^2 \Delta T_i$ and $\frac{\alpha_1 \beta}{\alpha_2} F(2\alpha_1, 2\alpha_2)$ with $\beta = \frac{r_1}{r_2}$ have the same first three moments.

This particular property allows to get C using the density of $F(2\alpha_1, 2\alpha_2)$:

$$f_{F(2\alpha_1, 2\alpha_2)}(x) = \frac{1}{B(\alpha_1, \alpha_2)} \left(\frac{\alpha_1}{\alpha_2}\right)^{\alpha_1} x^{\alpha_1-1} \left(1 + \frac{\alpha_1}{\alpha_2} x\right)^{-\alpha_1-\alpha_2} \quad (32)$$

In comparison to Monte Carlo results, it appears that Numerical integration and WoodF are the best, though WoodF struggle to have a convincing error term for maturities lower than 1 week.

Our current models have been showing good efficiency for small vol-of-vol values, which are what we have in local volatility, but struggle for high vol-of-vol values. To solve this, we will this time a Pure stochastic volatility Markov Switching volatility model to match the ATM volatilities. Here the spot FC rates are given by $\frac{dS_t}{S_t} = (r_d - r_f)dt + \sigma_{\xi(t)}dW_t$. We have lost the deterministic function A compared to the LSVMS model used before and therefore rely fully on the volatility linked to a Markov chain that defines different market conditions.

Consequently to the change in our model, we have now $C = \frac{E\left[\sqrt{\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 G_i^2 \Delta T_i}\right]}{E\left[\sqrt{\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 \Delta T_i}\right]}$.

We don't have a big change in the Lognormal mapping and WoodF methods after this change, in fact we just have to adapt de σ terms in the formulas. This small change is caused by the fact that they are moment matching methods and require at most the three first moments of $\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 G_i^2 \Delta T_i$ and $\sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 \Delta T_i$.

For Numerical Integration method, we have

$$\begin{aligned}
& E \left[\exp \left\{ -\psi \sum_{i=1}^N \sigma_{\xi(T_{i-1})}^2 G_i^2 \Delta T_i \right\} \right] \\
&= \sum_{1, \dots, i_{N-1}=1}^n \left(\prod_{j=0}^{N-1} \left(1 + 2\psi \sigma_{i_j}^2 \Delta T_{j+1} \right)^{-\frac{1}{2}} p_{i_j i_{j+1}}(T_j, T_{j+1}) \right) \quad (33)
\end{aligned}$$

with

$$p_{i,j}(s, t) = \mathbb{P}(\xi(t) = j | \xi(s) = i), s < t \text{ for any } i, j \text{ in } \{1, \dots, n\}.$$

Compared to the previous models, the Stochastic Volatility models developed here show a better fitting to Montecarlo results. The SV Numerical Integration method is more interesting from a theoretical point of view than really useful for real world pricing, due to the fact that evert step requires n^N terms to be calculated.

All these models can be adapted to price a volatility swap after its starting date. We therefore

$$\text{conclude with a new formula for } C, \text{ which is } C = \frac{E \left[\sqrt{m + \sum_{i=k}^N \sigma_i^2 G_i^2 \Delta T_i} \middle| \mathcal{F}_{T_{k-1}} \right]}{E \left[\sqrt{m + \sum_{i=k}^N \sigma_i^2 \Delta T_i} \middle| \mathcal{F}_{T_{k-1}} \right]}.$$

Here, m stands for the variance already observed on the market up to time T_{k-1} .

The model developed in this part is focused on the present. The problem we have tried to deal with is what the volatility is based on all the information we have up to date. It also focuses on short-term swaps, which differ from the Heston model and BN-S model presented before and presents a new approach.

To obtain satisfying results the model is based on the use of stochastic volatility, to register all the information possible from the data, and the use of Markov chains to simulate different market conditions. In the end, the model challenges were to estimate the correction term C, and we ended up with a formula (just above) that clearly highlights the fact that the model especially takes in account the past information (the term m and the filtration).

Data

We have described the different pricing models used for volatility swaps. Let's now focus on a practical approach using python based on the Heston model. Our objective is to follow a pricing process as we could do on a quantitative desk in order to price volatility swaps by making a comparison between different methods.

Calibration

We first decided to fetch data using yahoo finance on the S&P500 using 30D, 60D and 90D options. We used data plus or minus 15% of the moneyness with a strike at 5800. This data in a csv file is the foundation of our analysis throughout the rest of this document. The amount of data used is an important part of future challenges and decisions we had to take when doing the benchmark comparison.

We decided to use the Heston model as the base of our model. As such, we priced the different securities using the Black-Scholes formula : $C = S_0 N(d_1) - Ke^{-rT} N(d_2)$ with $d_1 = \frac{\ln(\frac{S_0}{K}) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$ (34) in order to obtain the prices of the options of our data for the S&P500. For the sake of simplicity, we decided to compute the implied volatility derived from these options using the bisection method. This allows for robustness and simplicity as the dataset was not overly complex. We then used the semi-closed form formulas described by Heston (1993) and the Fourier inversion of the characteristic function to find the price given by the pricing model. We implemented the Feller condition in the model given by Feller (1953): $2\kappa\theta \geq \sigma^2$ used to avoid the variance of any CIR process going to zero in order to avoid integration errors. While quantitative desks may let go of the general condition to have a model which fits the smile closely, we decided to keep this method for robustness.

We then tried to use different methods to minimize the loss function derived from our calibration: $[\text{Loss}(\theta) = \sum_{i=1}^N (C_{\text{model}}(K_i, T_i; \theta) - C_{\text{market}}(K_i, T_i))^2]$

We tried with both the Nelder-Mead and the L-BFGS to nuance the use of either one because of their different approach to minimization problems with the first one not needing the use of the gradient compared with the second one. In every case, we used the standard values for the different parameters of the Heston model as initial guess. By standard values we mean the values given by Dragulescu and Yakovenko (2002) which have been widely accepted as the standard for any benchmark using the Heston model:

$$\kappa = 11.35 \text{ (mean reversion speed)} \quad \theta = 0.022 \text{ (long - term variance)}$$

$$\sigma = 0.618 \text{ (volatility of variance)}$$

$$\rho = -0.64 \text{ (correlation between asset and variance)} \quad v_0 = 0.04 \text{ (initial variance)}$$

We then obtained this calibration:

$$\kappa = 5.9685, \theta = 0.0577, \sigma = 0.5494, \rho = -0.9037, v_0 = 0.0192$$

And these plots have been used to assess the quality of the calibration:

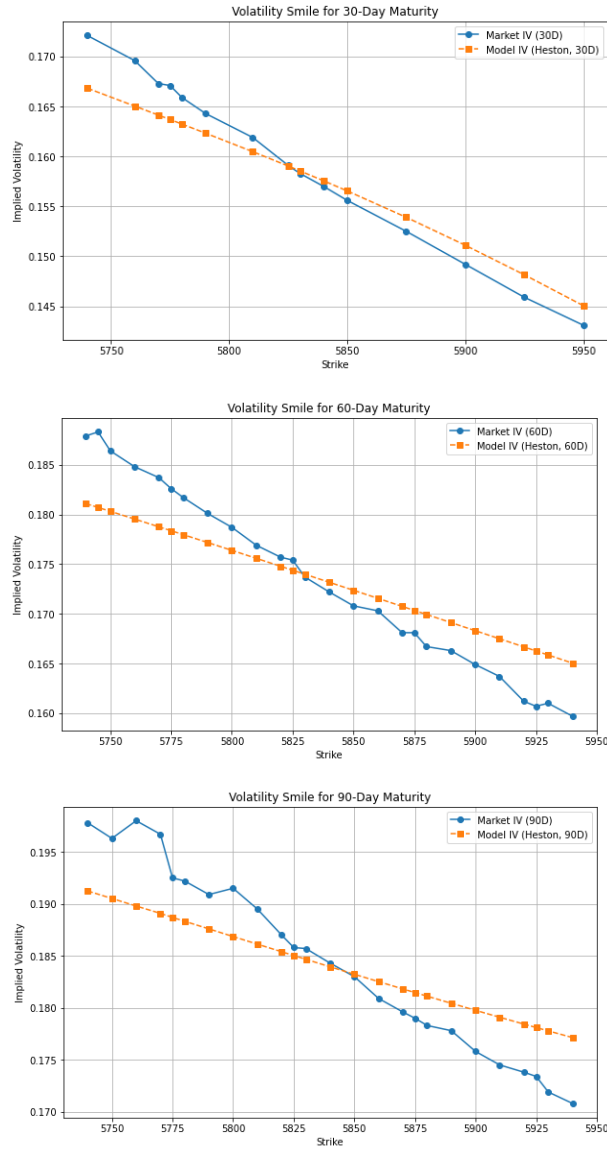


Figure 1: Graphs showing the calibration of our Heston model (in orange) compared to the actual implied volatility derived from our data (in blue) for the 30D, 60D and 90D options with strike and implied volatility as axes.

We notice that across the 3 types of maturities the calibrated model (in orange) follows the smile of volatility observed from our original data (in blue). The result is satisfactory across all maturities but with a higher precision for the 30-Day one. We thus concluded the first part of our data analysis with a fitted calibrated Heston model.

On a side note, here is the volatility surface deduced from our model. It clearly highlights what was said earlier on the shortcomings of the Black-Scholes model compared to the Heston model which allows us to capture the dynamics inherent to the smile.

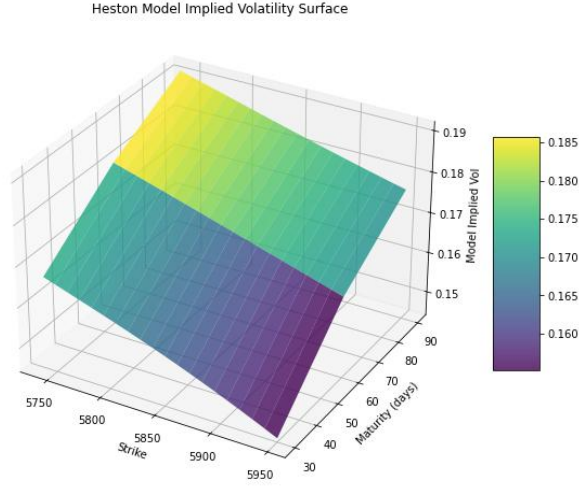


Figure 2: Implied volatility surface from our Heston model to demonstrate the limitation of the Black Scholes model with maturity, strike and implied volatility as axes.

Benchmarking

We then benchmarked different methods in order to assess the advantages and disadvantages of each one of them in pricing volatility swaps. We thus decided to compare the standard Monte Carlo method as our benchmark with a Monte Carlo analysis using a variance reduction technique and a random forest model. Our core idea was that Monte Carlo techniques are efficient but computationally expensive, so we decided to use the variance reduction technique to reduce this computational cost while experimenting to see if we could notice improvements. We also decided to add another model based on a random forest to test the rapidity of execution of a random forest (which is usually their principal application) even in a case of scarcity regarding the data. Based on this theoretical foundation for the control variate:

$$\widehat{V}_{CV} = \frac{1}{N} \sum_{i=1}^N [e^{-rT} \cdot \text{Payoff}_i + \varepsilon \cdot (E[\text{RealizedVol}] - \text{RealizedVol}_i)] \quad (35)$$

We then followed the recommendation of Rostan et al. (2012) and implemented the control variate with our epsilon as a truncated normal but between -2 and 2 for a better precision and a faster convergence. From the optimization process to find the best epsilon, we obtained the following plot :

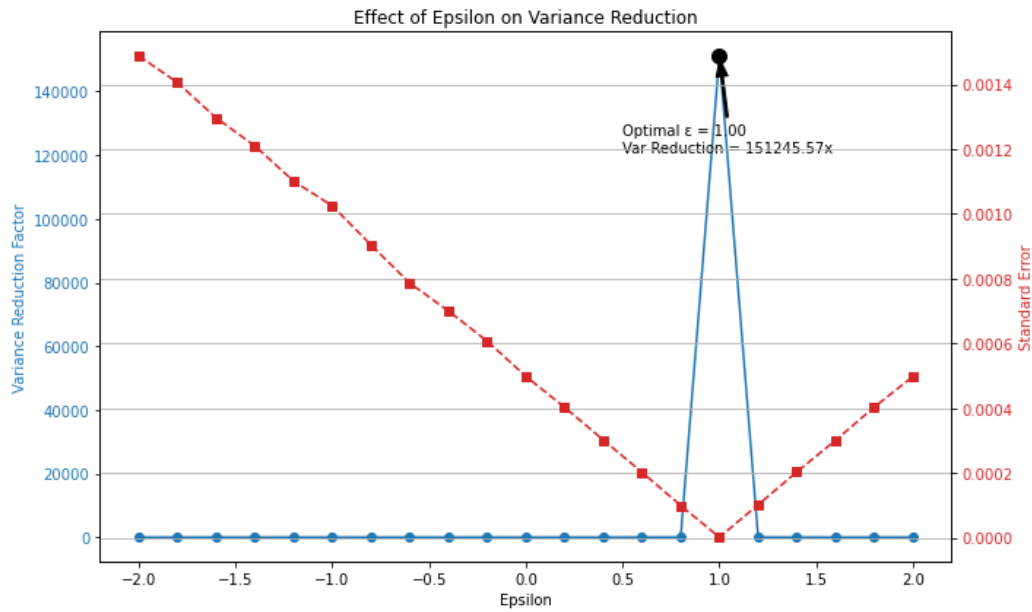


Figure 3: Variance reduction plot comparing the optimal epsilon with the variance reduction factor (in blue) and the associated standard error (in red).

We notice that the optimal epsilon is 1 and it induces a variance reduction factor of 151245.57 across all maturities observed in our dataset.

We assessed the gain out of this variance factor by comparing the Standard Monte Carlo with the Monte Carlo Control Variate under different metrics such as the Root Mean Square Error (RMSE), the efficiency (where we compare the computation time and the evolution in standard error) and the convergence rate.

We obtain the following results:

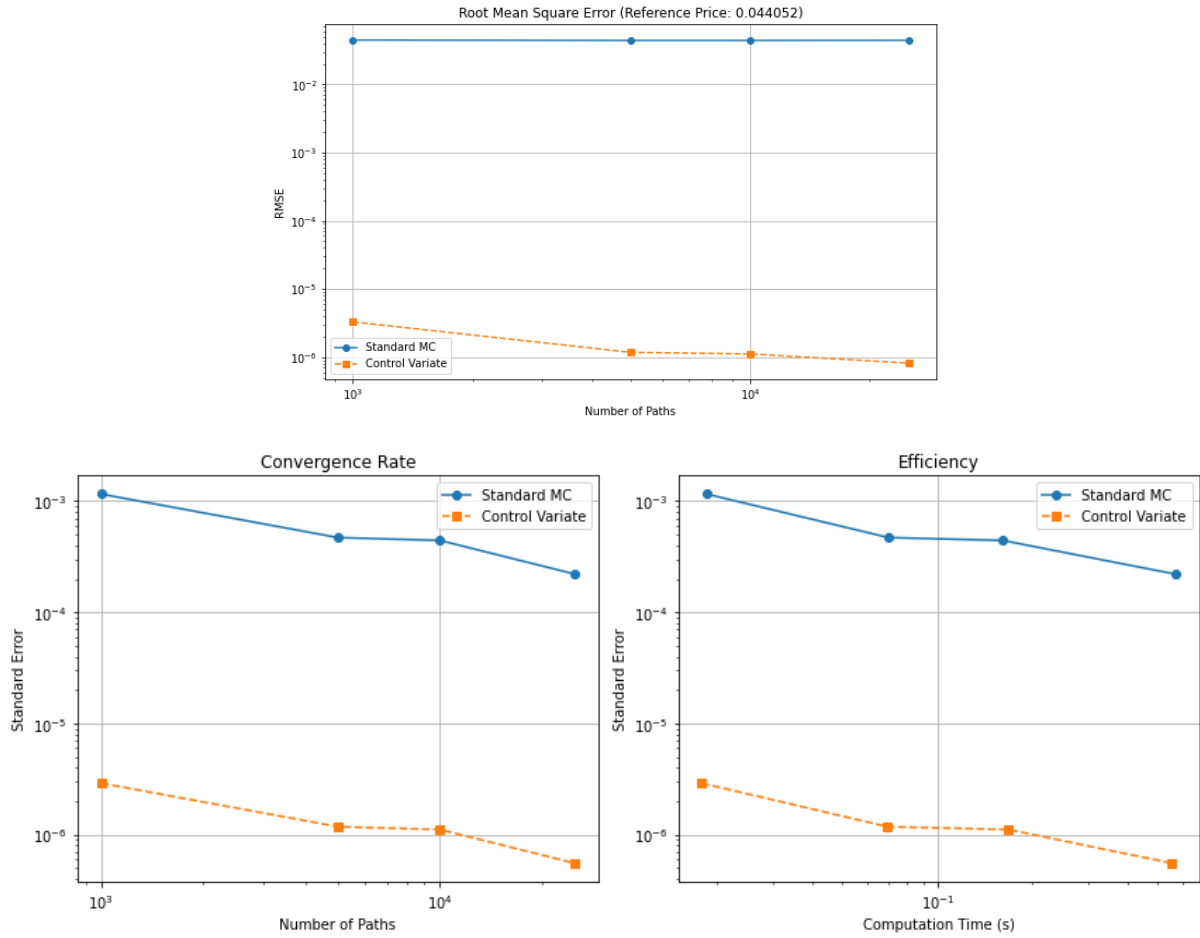


Figure 4: Comparison of the Standard MC (in blue) and the Control Variate (in orange) for different metrics (RMSE, standard error and compilation time)

Across all points of comparison, the Control Variate seems a better alternative to the standard Monte Carlo approach. Indeed, it drastically improves the quality of the prediction which is given by the RMSE (0.000001 against 0.01). But this also holds when we compare the rapidity of the convergence, and the efficiency compared to the computation time. This confirms what was analyzed by Rostan et al. (2012) in the fact that the control variate constitutes a better suited alternative both for reducing the computational cost but also in terms of precision.

We then pondered on a possible third option for a point of comparison to further or point about the advantages of the control variate method. We took inspiration from the work of Höcht, Schoutens and Verschuere (2024) in which they deepened the use of machine learning techniques to price capped volatility swaps with a focus on linear regression, random forest and gradient boosting techniques. While they built their model purely in a data driven manner, we decided to use our calibrated Heston model as a base for the random forest. While the authors highlighted the benefits of using a regression to price capped volatility swaps for the S&P500

compared to other stocks which benefited from random forests, we decided to confront this work once again. This train of thought emerged from different reasons, the first one is that contrary to their experiment, our model was based on a calibrated Heston model while their work was in a non-parametric setting, secondly, we are working with non-capped volatility swaps. We created a model based on a random forest using our initial Heston model and we decided to use shallow tree depth in order to avoid overfitting regarding the data at our disposition. Indeed, our data set contained less instances than the work of Höcht, Schoutens and Verschueren so we tried to avoid an overfitting result. We separated our data into both a training and a testing set and used cross validation to assess the quality of our parameters. As we said previously, the goal of our random forest was to give us a general idea of the precision of a faster model to serve as a base and contextualize the results obtained during our Monte Carlo analysis. As such, cross validation was performed with 5-fold and we minimize the loss function:

$$\underbrace{\text{CVLoss}(\theta) = \frac{1}{K} \sum_{k=1}^K L^{(k)}(\theta)}_{\text{Cross-validation loss over } K \text{ folds where}} \quad L^{(k)}(\theta) = \frac{1}{n_k} \sum_{i \in \mathcal{V}_k} l(y_i, \widehat{f^{(-k)}}(x_i; \theta)) \quad (36)$$

The objective of cross validation was to assess the quality of our parameters and not to find the optimal parameters through gridsearch because the scarcity of our data incentivized us to go with a broader setting for the random forest. From the cross validation, we extracted this output: CV RMSE: 0.004595 ± 0.001827 , Avg train time: 0.1247s. We can notice the rapidity of the training time, which is perfectly in line with what could be expected from a random forest as one of its main advantages is the ability to give faster results albeit less precise. Indeed, while the RMSE of 0.004595 is a rather satisfactory estimate, it is less precise than our previous Monte Carlo strategy and our Control variate method. To put this in perspective, we plotted different graphics to show the disparity between our naive random forest approach (in green) and the standard Monte Carlo technique (in blue) across the 30D, 60D and 90D maturity:

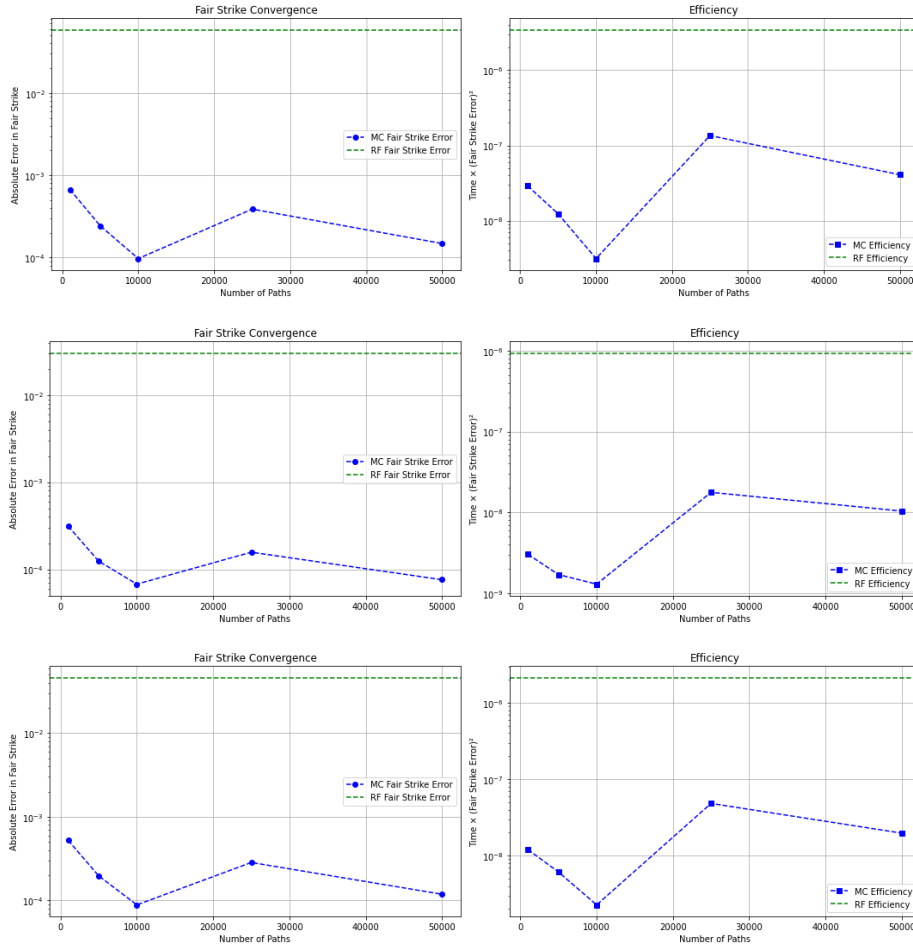


Figure 5: Comparison of the Standard MC (in blue) and the Random Forest (in green) for different metrics (RMSE, absolute fair strike convergence, efficiency)

This visual representation highlights what we described, the random forest is a quick and computationally cheap technique to produce a satisfactory result. Nonetheless, across all metrics and maturities the standard Monte Carlo approach has better efficiency and better fair strike convergence. We can notice the flat line represented by our random forest as the result of our pricing model because based on all the training done, the model gives one unique prediction for the pricing of volatility swaps.

To conclude the data analysis part, let's review what was discussed. From our analysis, the Control Variate method seems to be the most precise and efficient method across our benchmark. It avoids one of the principal disadvantages of the Monte Carlo method which is the computational power needed to provide optimal results in order to price volatility swaps. Moreover, when we put into perspective the results with the naive random forest approach, we

noticed clearly the expected theoretical results which are fast training but less precision. This comparison could of course benefit from further refinement for the random forest based on gridsearch to fine tune the parameters with a deeper tree depth based on a more extensive dataset in order to gain greater insights in a real-life scenario of random forest usage coupled with a walk forward technique like in the work of Höcht, Schoutens and Verschueren to make it dynamic.

Hedging

The hedging of volatility derivatives can be complex because of the nature of their payoff. Indeed, volatility swaps are delta neutral but vega sensitive and while delta hedging is often seen as a simple solution for most derivatives, it is not the case with these types of securities. The delta of volatility swaps is not linear and as such, other ways need to be considered for the hedging process. Three different options can be chosen to hedge volatility swaps: a strip of options, variance swaps or any volatility security using the greeks. To explain further the different methods, we need to define the vanna of a security which is defined as $Vanna: \frac{\partial^2 V}{\partial S \partial \sigma} = \frac{\partial Vega}{\partial S}$.

The first method consisting of using a strip of rebalanced options has been explored by different academicians. Broadie and Jain (2008) proposed the following formula to replicate a volatility swap under the Heston model:

$$V_e(0, T) = \frac{2}{T} \left[\int_0^T \frac{dS_t}{S_t} - \frac{S_T - S_0}{S_0} + \int_{S_0}^{\infty} \frac{1}{K^2} C_T(K) dK + \int_0^{S_0} \frac{1}{K^2} P_T(K) dK \right] \quad (37)$$

As we can see from the formula, the replication is operated with a long position in $1/K^2$ in call options from every maturity from S_0 to infinity and the same is true for a long position in put options from all the strike from 0 to S_0 as well as a short position in a forward contract at S_0 . Nonetheless, they give precision on the type of error stemming from this replication method. The first one comes from the rebalancing of the $1/S_t$ share of stock while the realized is computed in a discrete manner. The second one comes from the replication of the forward contract. Rolloos (2020) explored the same approach under Black-Scholes formulas: $d(V_{0,T}(t)T) \approx D(I^H \sqrt{t}) \approx N[dC^H(S, K_-) - \Delta^H(S, K_-) dS]$

and delta being $\Delta^H(S, K_-) = \Delta^{BS}(S, K_-, I^H) + v^{BS}(S, K_-, I^H) \frac{\partial I^H}{\partial S}$ by trading a delta hedged zero vanna option.

The second method is to use variance swaps in order to hedge our volatility swaps. Numerous relationships exist between the two securities. On a side note, there is an inequality between the fair strike of a variance swap and a volatility swap using Jensen's inequality which is: $K_{vol} \leq \sqrt{K_{var}}$. This has been a known inequality for a long time and highlights mathematically the

logical links which exist between variance and volatility. Broadie and Jain (2008) analyzed in greater detail this relationship and they demonstrated the existence of this hedge ratio $\beta_0 = (\partial X_t)/(\partial t) \cdot (\partial X_t)/(\partial t)$ where X denotes the price of a variance swap and Y a volatility swap. What it implies is that by constructing a portfolio of beta variance swap for each and every volatility swap held, the position is hedged perfectly regarding the volatility of the assets. In this article, they also expose the different relationships between the other greeks such as the mean reversion speed or the long run variance in an Heston model. All of the greeks they mention (delta, kappa, theta) always evolve through one type of equation which is $\frac{\partial \sqrt{X_t^T}}{\partial \text{greek}} = \frac{1}{2\sqrt{K_{\text{var}}}} \cdot \frac{\partial X_t^T}{\partial \text{greek}}$ (38) meaning that every aspect of the security can be hedged using a variance swap instead.

Lastly, a more general relationship exists which is only about the volatility of the asset. The objective of this type of equation is to express the derivative of the price based on volatility and to find a relationship with a number usually including volatility as well. From there, any volatility derivative matching this numerical value can be used to hedge the position. Such a formula can be found in the work of Elliott, Siu and Chan (2006). Nonetheless, the question of the other characteristic of the product not derived from volatility in the greeks remained unanswered in this paper.

Uses

- idea of dispersion trading

To understand this kind of strategies, where variance swaps can be very useful, we must enter in the modern portfolio theory, where returns on portfolios are a random variable. The risk of portfolio is measured by its variance (or standard deviation). The variance of a portfolio is

$$\sigma_{portfolio}^2 = \sum_{i=1}^n \omega_i^2 \sigma_i^2 + 2 \sum_{i=1}^n \sum_{j=i+1}^n \omega_i \omega_j \sigma_i \sigma_j \rho_{i,j}$$

Indices can be seen as portfolios of the assets they are composed of. For every index, we know the history of volatility, and the weights of the assets. We still need to compute the correlation between assets (ρ). The average index correlation, also called dispersion is

$$\bar{\rho} = \frac{\sigma_{index}^2 - \sum_{i=1}^n \omega_i^2 \sigma_i^2}{2 \sum_{i=1}^n \sum_{j=i+1}^n \omega_i \omega_j \sigma_i \sigma_j} \quad (39)$$

Dispersion trading consists in buying the volatility of the index constituents and selling the volatility of the index, which would be done using volatility swaps.

On finding that is important to evaluate the interest in this idea of strategy is that volatility is often over-valued by the markets – meaning, implied volatility is often higher than realized

volatility.

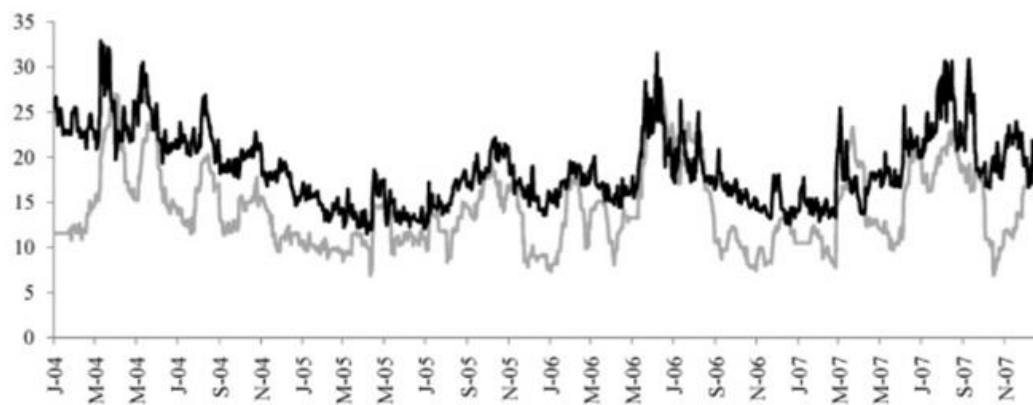


Figure 3.1: Implied (black) vs. realized volatility of DAX index

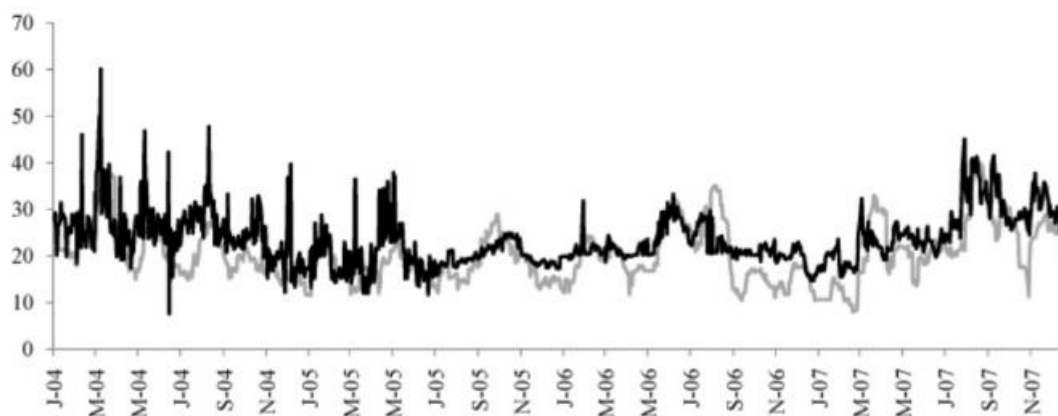


Figure 3.2: Implied (black) vs. realized volatility of Deutsche Bank AG

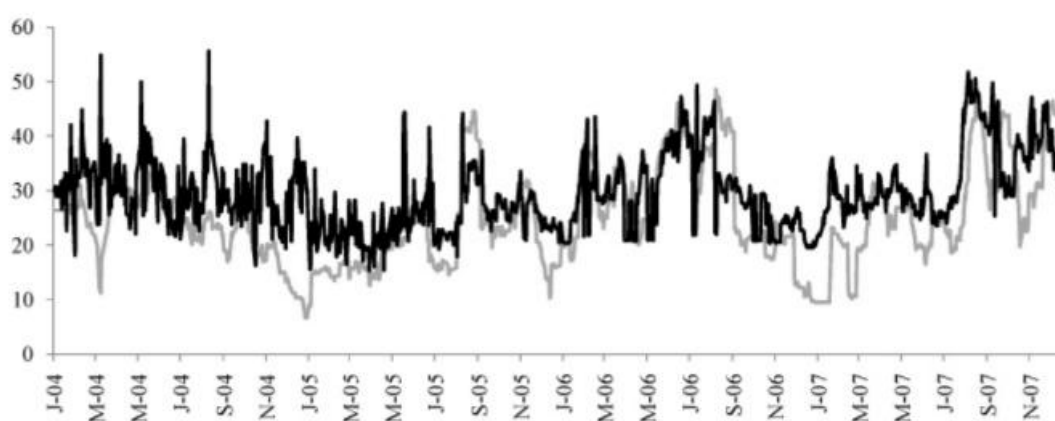


Figure 3.3: Implied (black) vs. realized volatility of Commerzbank AG

Sources: Silyakova.E (2009), Variance Swaps

The direct strategy consists in going short in index' variance and long constituent's variance, which on average generates positive payoff. The opposite move (short on constituents' variance, long on index' variance) is inverse dispersion strategy.

So we have the payoff of a direct dispersion strategy, which is a sum of variance swap payoffs of each constituent : $(\sigma_{R,i}^2 - K_{var,i}^2) * N_i$, where $K_{var,i}^2$ is the strike volatility of the swap, it's for the long leg of the strategy.

For the short leg (short position in index swap), we have $(K_{var,index}^2 - \sigma_{R,index}^2) * N_{index}$ where $N_i = N_{index} * \omega_i$.

Therefore, the payoff of our strategy is

$$N_{index} * \left(\sum_{i=1}^n \omega_i \sigma_{R,i}^2 - \sigma_{R,index}^2 - \sum_{i=1}^n \omega_i K_{var,i}^2 + K_{var,index}^2 \right) \quad (40)$$

We note that this kind of strategy can be difficult to implement perfectly in real life, just by considering the number of contracts necessary for some indices. Also, it requires to hold some positions that could be difficult to hold effectively : the $\sum_{i=1}^n \omega_i K_{var,i}^2 + K_{var,index}^2$ part of the payoff is from a replication strategy. This could generate important transaction costs and require human resources.

Conclusion

This paper has discussed and referenced the different pricing methodology and underlying models to price a volatility swap despite the few studies on the matter compared to other securities even among volatility derivatives. While the diversity in models exists, it is nonetheless recognized that general stochastic volatility models share theoretical foundations and thus similar results as we have discussed in this paper. The diversity in mathematical backgrounds is also complemented by a greater range of numerical implementations to obtain an actual price of the security. We have tried through this paper to propose a summary of all these different techniques. Throughout our research we have come to the conclusion that the discrete volatility setting seemed to be the most precise, albeit most difficult one to price. Monte Carlo as well as machine learning models could be a possibility to price these derivatives depending on whether we aimed for speed in the calculations or precision. Nonetheless, volatility swaps are still underused in our opinion as they constitute a precise tool to hedge against market movements and their uses in volatility trading. Perhaps focusing on their advantages as a tool on a trading desk and anchoring their usage to real life scenarios and the new volatile macroeconomic trend could fuel further interest and as such, academic research on how to price them.

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